12

Polarons' Seebeck coefficients

12.1 Definitions and general concepts

The Seebeck coefficient is defined as the open-circuit emf developed across a material in response to an imposed temperature differential: $\alpha \equiv \Delta\phi/\Delta T$. A general result of irreversible thermodynamics is that a material's Seebeck coefficient is also the entropy transported with a carrier during isothermal current flow divided by the carrier's charge q (Callen, 1960). As a result, the Peltier heat Π, the heat transported with a charge carrier during isothermal current flow, is related to the Seebeck coefficient by $\Pi \equiv qT\alpha$. The third law of thermodynamics ensures that Π and α both vanish as the temperature is lowered toward absolute zero.

The entropy transported with a carrier is the sum of two components (Emin, 2002). The first component only depends on the carrier's presence whereas the second component depends on the details of the mechanism by which it moves. In particular, the first component is the change of the material's entropy that results from adding a charge carrier. The second component is the net energy transfer required to move a carrier divided by the temperature. The first component generally contributes to polarons' Seebeck coefficients. The second component can contribute significantly when polarons move by phonon-assisted hopping.

Since a polaron's motion requires atomic motion a crystal's polaron bandwidth is generally very narrow, less than the characteristic phonon energy. Polarons' Seebeck coefficients are therefore first considered in the limit of vanishing polaron bandwidth. The Seebeck coefficient arising from adding a polaron of energy E and chemical potential μ_c is then $\alpha = (E - \mu_c)/qT$. Neglecting explicit consideration of polaron spins this Seebeck coefficient can be rewritten in terms of the fraction of the energy band's states that are occupied, $f = 1/\{\exp[(E - \mu_c)/\kappa T] + 1\}$ as $\alpha = (\kappa/q) \ln[(1 - f)/f]$. In this form it is evident that the narrow-band limit of the Seebeck coefficient changes sign at the band's half-filling ($f = ½$ and $E = \mu_c$). In these examples the entropy entering the Seebeck coefficient is recognized as just the

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change of the entropy-of-mixing S produced by adding a carrier to $n = fN$ existing carriers distributed among N states: $\alpha q = \partial S/\partial n$, where

$$S = -Nk[f \ln (f) + (1-f)\ln (1-f)]. \tag{12.1}$$

The spins of polarons' self-trapped electrons also contribute to their entropy and thereby to their Seebeck coefficients. In non-magnetic solids a self-trapped carrier's spin renders each individual polaron two-fold degenerate. As a result the polaron's Seebeck coefficient becomes $\alpha = (\kappa/q)\ln[2(1 - f)/f]$. By contrast, bipolarons produced as carriers self-trap as singlet pairs (1) generate no spin entropy, (2) double the electronic charge on each site and (3) halve the fraction of sites that are occupied to yield the Seebeck coefficient: $\alpha = (\kappa/2q)\ln[(2 - f)/f]$.

The two-fold spin degeneracy of a polaron with magnetic moment μ_B in a non-magnetic solid can be lifted with the application of a magnetic field H. The polaron's spin-entropy S_{sp} can be readily computed from the spins' free-energy:

$$S_{sp} \equiv \kappa\{\ln [2\cosh(\mu_B H/\kappa T)] - (\mu_B H/\kappa T)\tanh(\mu_B H/\kappa T)\}. \tag{12.2}$$

This spin entropy decreases from $\kappa\ln(2)$ at $H = 0$ to zero as $\mu_B H/\kappa T \rightarrow \infty$. This fall-off becomes significant for strong magnetic fields >10 T at low temperatures < 10 K when the polaron magnetic moment is taken as equal to that of a free electron. The polaron's magnetic moment can be determined by measuring the decrease of the Seebeck coefficient's spin entropy with increasing magnetic field. A zero magnetic moment implies that the carrier is actually a singlet bipolaron.

The energies of a semiconductor's electrons generally shift with temperature. These energy shifts result from two effects that are both proportional to the net phonon energy. Electronic energy levels shift because (1) atoms' equilibrium positions shift with changing temperature and (2) semiconductors' energy bands undergo phonon-mediated thermally induced broadening. These effects combine to generate the temperature dependence of a semiconductor's optical gap. The interactions that produce these effects could also affect carriers' Seebeck coefficients. The Seebeck coefficient arising from these interactions has been evaluated (Emin, 1977b; 1984; 1985):

$$\alpha = \frac{\left(\langle E + E_{ph}\rangle_1 - \langle E_{ph}\rangle_0 - \mu_c\right)}{qT}, \tag{12.3}$$

where μ_c denotes the chemical potential. The curved-bracketed expression is the net change of the energy of the system upon adding a charge carrier. The first contribution is the thermal average of the sum of a carrier's energy E, now a function of

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phonon frequencies and occupation numbers, and the phonon energy E_{ph}. The subscript 1 indicates that the thermal average is performed with vibration frequencies that have been modified by the added carrier's presence. The second contribution within the curved bracket denotes the thermal average of the phonon energy in the absence of the carrier (denoted by the subscript 0). As indicated by optical measurements the thermal average of a carrier's electronic energy $\langle E \rangle_1$ shifts with temperature. The carrier-induced change of the thermal average of the vibration energy $\langle E_{\mathrm{ph}} \rangle_1 - \langle E_{\mathrm{ph}} \rangle_0$ is also temperature dependent. The temperature dependences of both terms become strongest in the classical limit where κT exceeds the phonon energies. However when these two temperature dependences become strongest they also cancel one another. In other words the terms within the curved brackets show no net temperature dependence within the high-temperature regime. Thus thermal expansion and thermally induced phonon-mediated broadening of a semiconductor's energy bands do not significantly affect the curved-bracketed expression.

The narrow-band limit fails when the thermal energy κT falls below the bandwidth. Beyond the narrow-band limit a Boltzmann-equation-based analysis finds the Seebeck coefficient to be an average of the Seebeck coefficients from states of different energy weighted by $\sigma(E)$, each energy's contribution to the net electrical conductivity:

$$\alpha = \frac{1}{qT} \frac{\int dE \sigma(E)(E - \mu_c)}{\int dE \sigma(E)} \,. \tag{12.4}$$

Indeed this average extends over all energy bands that contribute to a semiconductor's electrical conductivity. Generally contributions from nearly empty conduction-band states are negative and those from nearly filled valence-band states are positive. As the temperature is lowered the chemical potential tends to become lodged among partially occupied states that reside within a semiconductor's intrinsic energy gap. With the electrical conductivity in the low-temperature limit presumed to be dominated by transport among states that are within κT of μ_c, the third law of thermodynamics is satisfied since then $\alpha \to 0$ as $T \to 0$.

12.2 Hopping polarons' Seebeck coefficients

The Seebeck coefficient has been calculated for carriers that move by phonon-assisted hopping. In particular, the open-circuit emf has been computed for a carrier that hops between sites that are each assumed to be coupled to independent phonon baths held at different temperatures (Emin, 1975b). The Seebeck coefficient for a carrier of charge q hopping between sites i and j depends on the energies of equilibrated carriers at these sites, E_i and E_j, and on measures of these sites' electron–phonon coupling strengths, Γ_i and Γ_j :

$$\alpha_{i,j} = \frac{E_i\left(\frac{\Gamma_j}{\Gamma_i + \Gamma_j}\right) + E_j\left(\frac{\Gamma_i}{\Gamma_i + \Gamma_j}\right) - \mu_c}{qT}, \tag{12.5}$$

where T denotes the mean temperature of the two sites and μ_c signifies the carrier's chemical potential. In the semiclassical limit in which atoms vibrate classically, $\Gamma_i/(\Gamma_i + \Gamma_j) = E_{p,i}/(E_{p,i} + E_{p,j})$, where $E_{p,i}$ and $E_{p,j}$ denote the corresponding sites' polaron binding energies. A site's polaron binding energy is the net energy lowering produced by atomic displacements when a carrier occupies the site. As such, a site's polaron binding energy contributes $-E_{p,i}$ to its net energy E_i. The polaron binding energy of a localized state generally decreases as its spatial extent is increased.

For hopping between equivalent ($\Gamma_i = \Gamma_j$) iso-energetic ($E_i = E_j = E$) sites the Seebeck coefficient becomes $\alpha_{i,j} = (E - \mu_c)/qT$. The Seebeck coefficient's characteristic energy $E_\alpha \equiv E - \mu_c$ is then just the activation energy for generating carriers of energy E in the narrow-band limit of a semiconductor. As such, at sufficiently high temperatures E_α differs from the activation energy of the crystal's dc electrical conductivity E_σ by the activation energy of a hopping polaron's semiclassical mobility, $E_\mu \equiv E_\sigma - E_\alpha$. For polaron hopping the semiclassical mobility's activation energy generally exceeds the characteristic phonon energy. Thus, polaron hopping is indicated if the electrical conductivity's activation energy E_σ exceeds E_α by at least the energy of a typical phonon.

When hopping occurs between non-degenerate ($E_i \neq E_j$) states whose electron–phonon couplings are inequivalent ($\Gamma_i \neq \Gamma_j$) some vibrational energy is transferred with the carrier as it hops. For example, consider hopping between a strongly coupled state of energy E_s with electron–phonon coupling Γ_s and a weakly coupled state of energy E_w with electron–phonon coupling Γ_w. Then with $\Gamma_s \gg \Gamma_w$ the Seebeck coefficient of Eq. (12.5) becomes $\alpha_{i,j} = (E_w - \mu_c)/qT$. That is, the energy which enters into the Seebeck coefficient is that of the state whose electron–phonon coupling is weakest. A localized electron's energy and coupling to phonons tends to increase with the severity of its localization. As illustrated in Fig. 12.1 a hop from a strong-coupling state to the higher energy weak-coupling state occurs with the absorption of vibration energy $\Delta \equiv E_w - E_s$ at the strong-coupling site. Similarly a hop from the weak-coupling state to the strong-coupling state occurs with the emission of vibration energy at the strong-coupling site. That is, the vibration energy Δ is transferred with the carrier as it hops.

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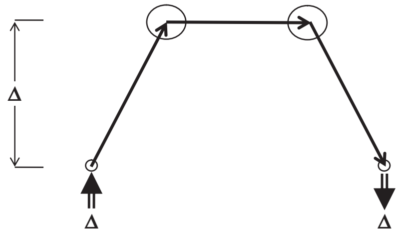

Fig. 12.1 Three hops are represented by solid arrows. The upward-directed double-lined arrow depicts absorption of vibration energy Δ at a severely localized state (small circle) facilitating a phonon-assisted hop to a weakly localized state (large circle). The horizontal solid arrow indicates a jump between equivalent degenerate weakly localized states. The downward-directed double-lined arrow depicts emission of vibration energy Δ at a severely localized state enabling a phonon-assisted hop from a weakly localized state.

In summary no vibration energy is transferred with direct polaron hops between equivalent states. As a result such polaron hopping yields a characteristic Seebeck energy that is less than the conductivity activation energy. By contrast, vibration energy is transferred with a carrier as it hops between non-degenerate inequivalent states. With this energy-transfer contribution the Seebeck energy becomes tied to the more weakly coupled state. Thus Seebeck coefficients of carriers moving among weak-coupling states are not affected by capture and release at strong-coupling traps. The conductivity activation energy and the characteristic Seebeck energy are then both associated with the weak-coupling states. Therefore comparison of the conductivity activation energy and the Seebeck energy provides a means of distinguishing between polaron hopping and trap-modulated (e.g. bandtail) conduction.

12.3 Softening (bi)polarons' Seebeck coefficients

Dynamic shifts of a localized carrier's charge distribution in response to atoms' vibrations generally alter their frequencies. In particular, the ground-state energy of a localized carrier is reduced as its centroid, size and/or shape adjust to the slowly modulating potential produced by vibrations of surrounding atoms. The localized carrier's relaxation thereby lowers the net energy and stiffness constants associated with these atoms' vibrational displacements. When a ground-state carrier's adjustment to atomic movement is treated by perturbation theory the lowered stiffness constants are given by Eq. (4.15) with the electron–phonon interaction inducing virtual transitions of the carrier from its ground state to its excited states.

As illustrated in Section 3.3, carrier-induced softening increases with the strength of the electron–phonon interaction until its matrix elements become large enough compared with the inter-level energy separations to induce a shift

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of atoms' equilibrium positions. Indeed, symmetry-breaking atomic displacements linking electronic states of different symmetry are maximally softened when their energy separation is just large enough to forestall the Jahn–Teller effect's symmetry-breaking shift of atoms' equilibrium positions (Jahn and Teller, 1937; Öpik and Pryce, 1957). A similar phenomenon affects a singlet bipolaron formed from degenerate orbital states. In particular, as discussed in Section 7.3, symmetry-breaking atomic displacements linking a singlet bipolaron's two-carrier states with different symmetries are most strongly softened when the difference of their exchange energies is just large enough to prevent symmetry-breaking shifts of atoms' equilibrium positions (Emin, 2000b). All told, carrier-induced softening becomes most significant when the carrier's electronic states are easily altered. Carrier's electronic states are exceptionally mutable when they are nearly degenerate or are large and particularly polarizable [cf. Eq. (4.18)].

Carrier-induced softening by a static polaron is absent from models whose simplifications preclude its localized carrier's wavefunction adjusting to movements of surrounding atoms. In particular, there is no carrier-induced softening for a stationary small polaron within Holstein's molecular-crystal model (Holstein, 1959b). Its small polaron is only characterized by shifts of the equilibrium values of its molecules' deformation coordinates.

Nonetheless, a softened region surrounding a localized carrier generally scatters phonons thereby increasing their thermal resistivity. Carrier-induced softening may even foster phonons' localization (Emin, 1990; 1991b). Moreover, carrier-induced softening enhances carriers' Seebeck coefficients via two effects (Emin, 1999). First, adding a charge carrier thereby (1) increases the entropy of atoms' vibrations and (2) augments the transfer of vibration energy associated with a polaron's hopping.

To illustrate these two effects consider polaron motion among molecules whose harmonic vibrations are softened upon occupation by a charge carrier. The entropy of an harmonic oscillator having angular frequency ω_i is $S_i = -\partial F_i/\partial T$, where the free-energy of the oscillator is $F_i = \kappa T \ln[2 \sinh(\hbar\omega_i/2\kappa T)]$. The first-order contribution to the Seebeck coefficient from a set of frequency shifts that are each denoted by $\Delta\omega_i$ is:

$$\alpha_{\mathrm{vib}} = \frac{1}{q}\sum_i \frac{\partial S_i}{\partial \omega_i}\Delta\omega_i = \frac{\kappa}{q}\sum_i \left(\frac{-\Delta\omega_i}{\omega_i}\right)\left[\frac{(\hbar\omega_i/2\kappa T)}{\sinh(\hbar\omega_i/2\kappa T)}\right]^2. \tag{12.6}$$

As the temperature is raised from zero the square-bracketed expression increases monotonically from zero toward unity as $\hbar\omega_i/2\kappa T \to 0$.

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Carrier-induced softening enhances the transfer of vibration energy with a thermally assisted hop. In particular, carrier-induced stiffness changes enable vibration energy to be transported with a carrier as it hops between equivalent sites. By contrast, in the absence of carrier-induced stiffness changes the transfer of vibration energy during a thermally assisted hop only occurs for hops between sites with inequivalent couplings to the phonons, cf. Section 12.2.

An example illustrates how a carrier-induced change of stiffness affects the transport of vibration energy. Consider the net harmonic deformation energy of two sites whose deformation coordinates are x_1 and x_2. A carrier's occupation of site 1 causes its deformation coordinate to be displaced to F/k_c as its stiffness constant is shifted from k to k_c:

$$E_{\text{def}} = \frac{k_c}{2}(x_1 - F/k_c)^2 + \frac{k}{2}x_2^2, \tag{12.7}$$

where F is the force associated with the electron–phonon interaction. A semiclassical hop between equivalent sites can occur when their electronic energies come into coincidence, $W(x_1) = W(x_2)$ at $x_1 = x_2$. The minimum deformation energy needed to establish such a coincidence occurs when $x_1 = x_2 = F/(k_c + k)$. The deformation energies at sites 1 and 2 associated with establishing this minimum energy coincidence are $F^2[k^2/k_c(k + k_c)^2]/2$ and $F^2[k/(k + k_c)^2]/2$, respectively. The sum of these two energies is the activation energy for the semiclassical polaron hop $E_b[k/(k + k_c)]$, where $E_b \equiv F^2/2k_c$ is the polaron's binding energy. The difference between the two sites' deformation energies at the minimum-energy coincidence characterizes the vibration energy transported in a hop, $E_b[k/(k + k_c)][(k - k_c)/(k + k_c)]$. To first order in the carrier-induced shift of the local vibration frequency $\Delta\omega \equiv \omega_c - \omega$ the transported energy becomes $(E_b/2)(-\Delta\omega/\omega)$, where the stiffness k is related to the vibration frequency ω by $k \propto \omega^2$.

This transported vibration energy divided by qT contributes to the Seebeck coefficient for high-temperature semiclassical polaron hopping. As the temperature is lowered below the high-temperature semiclassical limit the hopping activation energy and the vibration energy transported with a hop both decline. Indeed a general treatment of the vibration energy contribution to the Seebeck coefficient yields (Emin, 1999):

$$\alpha_{\text{trans}} = \frac{\kappa}{q}\sum_i \left(\frac{-\Delta\omega_i}{\omega_i}\right)\left(\frac{E_{b,i}}{2\hbar\omega_i}\right)\left[\frac{(\hbar\omega_i/\kappa T)^2}{\sinh(\hbar\omega_i/\kappa T)}\right] \tag{12.8}$$

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for carrier-induced softening between equivalent sites, where $E_{b,i}$ is the contribution to the polaron binding energy from the i-th vibration mode. The square-bracketed factor in Eq. (12.8) peaks at $T \approx \hbar\omega_i/2\kappa$ while it vanishes as $T \to 0$ and as $T \to \infty$.

The enhancement of the Seebeck coefficient from carrier-induced softening is the sum of α_{vib} and α_{trans}. The temperature dependence of each of these contributions is governed by those of the square-bracketed factors contained within its i-summation. These factors, respectively defined as $f_{\text{vib},i}$ and $f_{\text{trans},i}$, are plotted in Fig. 12.2.

The magnitude of α_{trans} can be crudely underestimated by setting $E_{b,i}/2\hbar\omega_i = 1$. Carrier-induced softening's contribution to the Seebeck coefficient is then roughly κ/q multiplied by the product of the fractional softening $\Delta\omega/\omega$ and the number of softened modes. Typical carrier-induced softening is about 10% (Austin and Mott, 1969). Then each softened mode contributes ~10 μV/K to the net Seebeck coefficient.

However, as described at the beginning of this section, carrier-induced softening sometimes can be especially large. The unexpectedly large Seebeck coefficients in boron carbides have been attributed to especially large carrier-induced softening (Aselage *et al.*, 1998; 2001). This exceptional softening has been ascribed to the formation of singlet bipolarons amongst the nearly degenerate frontier orbitals of boron carbides' icosahedra (Emin, 2000b). A significant contribution to pentacene's Seebeck coefficient has also been attributed to carrier-induced stiffness changes (von Mühlenen *et al.*, 2007).

Semiconductors whose carriers yield unusually large Seebeck coefficients while also bolstering a material's thermal resistivity ρ_T can be useful in thermoelectric devices. Indeed, both effects enhance a material's thermoelectric figure-of-merit $\alpha^2\sigma\rho_T$, where σ denotes the electric conductivity. To this end pursuing materials that might have exceptionally large carrier-induced stiffness changes provides an avenue

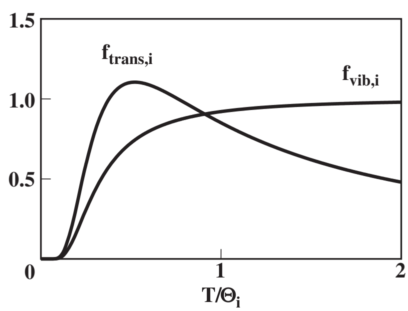

Fig. 12.2 The temperature-dependent factors $f_{\text{trans},i}$ and $f_{\text{vib},i}$ are plotted against T/Θ_i, where $\Theta_i \equiv \hbar\omega_i/\kappa$.

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for finding promising materials for thermoelectric applications. For example, boron carbides' exceptional thermoelectric properties combined with the huge neutron absorption cross-section of ^{10}B provides the basis of a solid-state neutron detector (Emin and Aselage, 2005).

12.4 Seebeck coefficients of polarons in magnetic semiconductors

Polarons are reported in narrow-band magnetic materials including some transition-metal and rare-earth oxides. These materials' magnetism is attributed to local magnetic moments primarily from unpaired spins of electrons localized on cation sites. Intra-site exchange interactions foster alignment of each site's unpaired spins. In addition, comparatively weak inter-site exchange effects promote ordering of local moments' relative orientations. Super-exchange interactions involving the nominally non-magnetic ligands that reside between magnetic ions generate the antiferromagnetic orderings which characterize most magnetic insulators (Kramers, 1934; Anderson, 1950). The prototypical illustration is MnO, an antiferromagnetic insulator with a spin of 5/2 on each of its Mn^{2+} cations. Nonetheless direct exchange between magnetic ions sometimes generates ferromagnetic ordering. The classic example is EuO, a ferromagnetic insulator with a 7/2 spin on each of its Eu^{2+} cations (Methfessel and Mattis, 1968).

Adding a severely localized charge carrier to a magnetic site alters its magnetic moment and the associated exchange interactions. These changes alter the orientations available to the localized electrons. The magnetic system's entropy is thereby altered by the addition of a charge carrier. The carrier-induced change of the magnetic entropy divided by the carrier's charge provides the magnetic contribution to the carrier's Seebeck coefficient (Liu and Emin, 1984).

Local moments are well ordered at temperatures well below the magnetic ordering temperature T_m. Then the magnetic contribution to a carrier's Seebeck coefficient at temperature T is:

$$\alpha_{mag} \cong \mp\frac{\kappa}{q}\left[\left(\frac{T_m}{T}\right)\exp\left(-\frac{T_m}{T}\right)\right], \tag{12.9}$$

where the upper and lower signs respectively pertain for a carrier increasing or decreasing the occupied site's magnetic moment (Liu and Emin, 1984). The magnitude of α_{mag} is very small at low temperatures and increases as the temperature is raised toward the magnetic ordering temperature.

At much higher temperatures $T > T_m$ the magnet enters its paramagnetic regime. As the temperature is increased constraints on the orientations of local moments are

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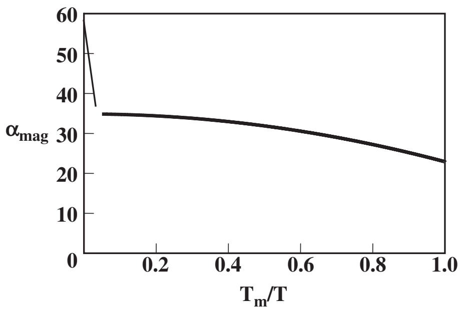

Fig. 12.3 The magnetic contribution to the Seebeck coefficient in the paramagnetic regime $T > T_m$ is plotted against T_m/T. The dark line is obtained from Eq. (12.10) with $S_c = 1$, $S = ½$ and $A_{ex} = 5/36$. The thin curve indicates the behavior as the thermal energy is raised above the intra-site exchange energy.

reduced and they become free to rotate independently. The magnetic contribution to the Seebeck coefficient then takes the form (Liu and Emin, 1984)

$$\alpha_{mag} \cong \frac{\kappa}{q}\left[\ln\left(\frac{2S_c+1}{2S+1}\right) \mp A_{ex}\left(\frac{T_m}{T}\right)^2\right], \tag{12.10}$$

where S_c and S are the values of the local moment's spin with and without the carrier's presence. The corresponding multiplicities of the local moment's spin state are $2S_c + 1$ and $2S + 1$. Again the upper and lower signs respectively pertain when $S_c > S$ and $S_c < S$ with A_{ex} denoting a numerical constant that is of the order of unity. In the extreme paramagnetic limit $T \gg T_m$ inter-site exchange becomes ignorable. Then only the first term within the square brackets of Eq. (12.10) survives (Heikes *et al.*, 1963). For large spins $S \gg 1$, $\alpha_{mag} \to \pm(\kappa/q)[1/(2S + 1)]$, where the sign indicates whether the addition of a carrier increases or decreases the site's spin.

Finally, in the generally unattainable ultra-high-temperature limit the thermal energy rises well above the intra-site exchange energy. In this regime even intra-site exchange effects can be ignored. Since the carrier is then unaffected by exchange interactions, the magnetic contribution to its Seebeck coefficient reverts to that for a spin-1/2 carrier in a non-magnetic solid: $\alpha_{mag} \to \alpha_{sp}$ of Eq. (12.2).

Figure 12.3 illustrates the results of a model calculation of α_{mag} in the paramagnetic regime (Liu and Emin, 1984). Two regions with different temperature dependences are encountered as the temperature is raised above T_m. These two regions correspond to the carrier's inter-site and intra-site exchange interactions being successively overwhelmed. The carrier-related spin entropy rises as the constraints imposed by these exchange interactions are progressively overcome.